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1.4 Pre-Quiz: Arithmetic Sequences

Homework. Due Monday 21 September, at the start of class. Write your answers in the boxes; a calculator is allowed. Several parts ask you to write a sentence — those sentences are the point of the assignment.

1. Here are five sequences.

- (i) 6, 10, 14, 18, 22, ...
- (ii) 3, 6, 12, 24, 48, ...
- (iii) 1, 4, 9, 16, 25, ...
- (iv) 20, 14, 8, 2, -4, ...
- (v) 5, 5, 5, 5, 5, ...

- (a) For each sequence, write down the three differences $u_2 - u_1$, $u_3 - u_2$ and $u_4 - u_3$.
- (b) State which of the five sequences are arithmetic, and write down the common difference d of each one.
- (c) For each sequence that is *not* arithmetic, describe its rule in one sentence.
- (d) Complete the sentence: “A sequence is arithmetic when ...”

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2. Two words in this unit sound alike but mean different things: *sequence* and *series*.

- (a) Explain the difference between a sequence and a series, in your own words.
- (b) Consider the sequence 4, 9, 14, 19, 24. Write down the series formed from these five terms, and find its value.
- (c) For that sequence, state what u_5 means and what S_5 means, and write down the value of each.
- (d) State whether each of the following is a sequence or a series:

3, 7, 11, 15

$3 + 7 + 11 + 15$

$u_n = 2n + 1$

$S_{20} = 430$

- (e) Explain why $S_5 - S_4 = u_5$.

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3. When Carl Friedrich Gauss was a schoolboy, his teacher asked the class to add all the whole numbers from 1 to 100. Gauss wrote down the answer almost at once.

(a) Find the value of $1 + 100$, of $2 + 99$, and of $3 + 98$. Write down what you notice.

(b) How many pairs like these are there in $1 + 2 + 3 + \cdots + 100$? Hence find the total.

(c) The sum $1 + 2 + 3 + \cdots + 100$ is an arithmetic series. Write down u_1 , d and u_{100} , then use $S_n = \frac{n}{2}(u_1 + u_n)$ to check your answer to (b).

(d) Explain, in one or two sentences, why Gauss's pairing and the formula are the same idea. Where does the $\frac{n}{2}$ come from?

(e) Use Gauss's method to find $1 + 2 + 3 + \cdots + 500$.

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4. The *Fibonacci* sequence begins

$$1, 1, 2, 3, 5, 8, 13, \dots$$

Each term after the second is the sum of the two terms before it.

- Write down the next three terms.
- Show that the Fibonacci sequence is not arithmetic.
- The rule $u_{n+2} = u_{n+1} + u_n$, with $u_1 = u_2 = 1$, is called a **recursive** rule. Explain what makes a rule recursive.
- The arithmetic sequence 7, 11, 15, 19, ... can be written recursively as $u_1 = 7$, $u_{n+1} = u_n + 4$, or explicitly as $u_n = 4n + 3$. Use each rule to find u_5 . Then state which of the two rules you would use to find u_{100} , and why.
- The ninth to twelfth Fibonacci terms are 34, 55, 89, 144. Use your calculator to find $\frac{u_{10}}{u_9}$ and $\frac{u_{12}}{u_{11}}$, and write down what you notice.

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5. In the news about artificial intelligence you will hear the phrase **recursive self-improvement** (RSI): the idea that an AI system could improve its own design, and that the improved system could then improve itself again, round after round.

Here are two simple models of a system's capability score c_n after n rounds. Both start at $c_1 = 100$.

Model A: each round adds 20 points.

Model B: each round adds 20% of the current score.

- (a) Using your answer to Question 4(c), explain what the word *recursive* is doing in the phrase “recursive self-improvement”.
- (b) Write down the first four terms of each model. State which model is arithmetic and give a reason.
- (c) Write a recursive rule for each model.
- (d) Find c_{10} for each model.
- (e) People who worry about RSI have in mind a process that speeds up as it goes. State which model matches that description, and explain what feature of the model makes it speed up.

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6. Consider the arithmetic series

$$4 + 11 + 18 + \cdots + 200.$$

- Write down the value of the common difference d .
- Find the number of terms in the series.
- Find the sum of the series.
- An arithmetic sequence has $u_1 = 6$ and $d = -2$. Find S_{30} .
- An arithmetic series has $u_1 = 5$ and $S_{10} = 320$. Find the value of the common difference d .